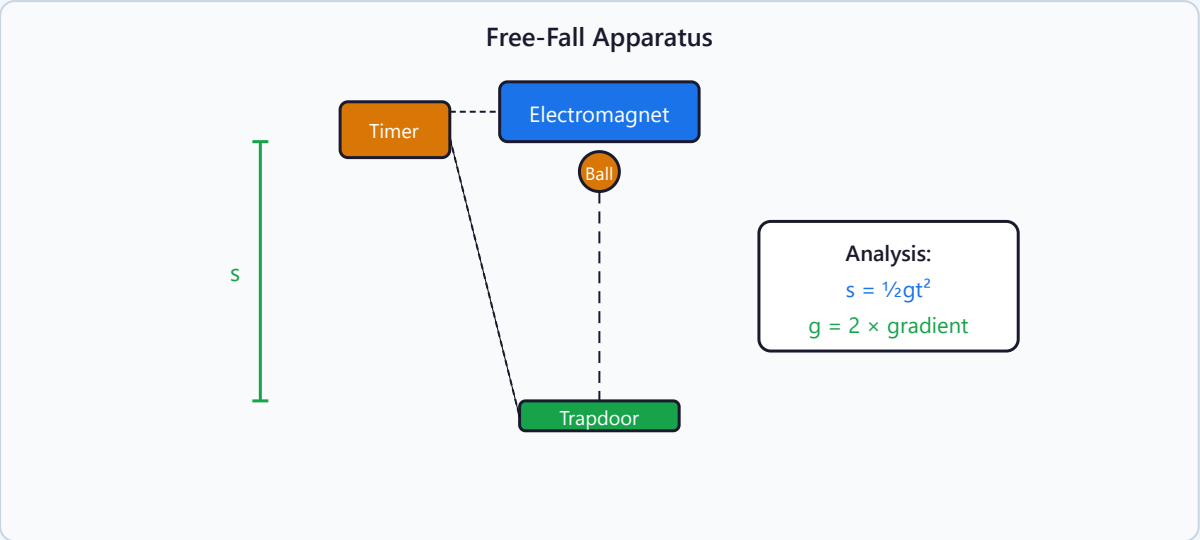
RAG Learning

OCR A A-Level Physics — Practical Activity Groups (PAGs)

Table of Contents

1. PAG1: Determination of g by Free-Fall
2. PAG2: Stationary Waves on a String
3. PAG3: Resistivity of a Wire & V-I Characteristics
4. PAG4: Young Modulus of a Wire
5. PAG5: Inverse-Square Law & Absorption of Radiation
6. PAG6: Capacitor Charge and Discharge
7. PAG7: Investigation Using Data Analysis
8. PAG8: SHM (Mass-Spring and Pendulum)
9. PAG9: Specific Heat Capacity
10. PAG10: Force on a Current-Carrying Wire
11. PAG11: EMF and Internal Resistance
12. PAG12: Photoelectric Effect Investigation

⚗ PAG1: Determination of g by Free-Fall



Apparatus

- Electromagnet with release switch
- Steel ball bearing
- Trapdoor switch and electronic timer
- 2 m ruler, clamp stand

Method

1. Clamp electromagnet at height s above the trapdoor.
2. Release ball; timer starts when ball releases and stops when trapdoor opens.
3. Record t. Repeat 3 times at each height; take mean.
4. Vary s over at least 6 values (0.2–1.0 m).
5. Plot s vs t^2. Gradient = g/2.

Safety

- Soft pad beneath trapdoor
- Stable clamp stand; feet clear of falling ball

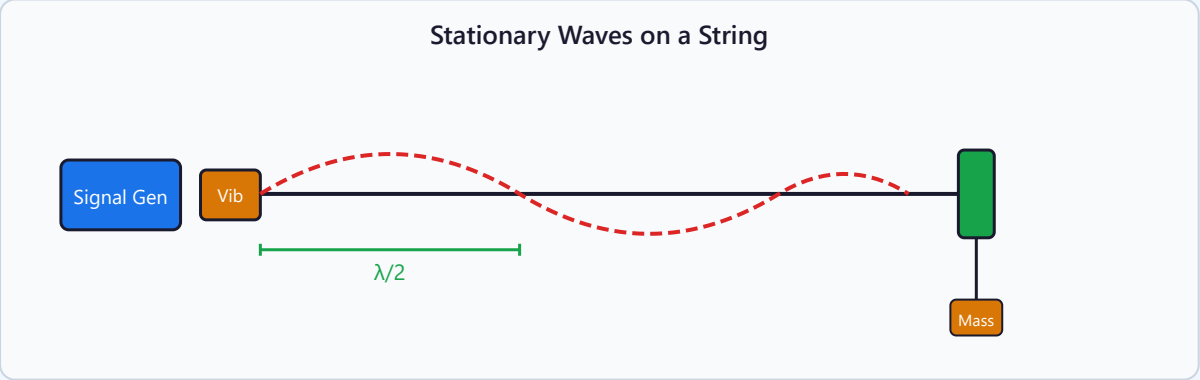
Analysis

$$s = \tfrac{1}{2}gt^2 \Rightarrow \text{gradient of s vs } t^2 = g/2$$

Source of Error	Type	Improvement
Residual magnetism delays release	Systematic	Use demagnetising coil or mechanical release
Air resistance	Systematic	Use small, dense ball
Height measurement	Random	Measure from bottom of ball to trapdoor centre
Timer resolution	Random	Use high-resolution electronic timer

Exam Tip: The graph should pass through the origin. A positive y-intercept suggests a timing delay (systematic error). Calculate percentage difference from accepted g = 9.81 m s^{-2} to evaluate accuracy.

⚗ PAG2: Stationary Waves on a String



Apparatus

- Signal generator and vibrator
- String, pulley, bench clamp
- Slotted masses, metre ruler

Method

1. Attach string to vibrator, pass over pulley with hanging mass for tension.
2. Adjust frequency for fundamental mode. Record f and measure L.
3. Find higher harmonics. Record frequency and number of loops.
4. Vary tension or string type and repeat.

Safety

- Safety screen; soft pad beneath masses

Analysis

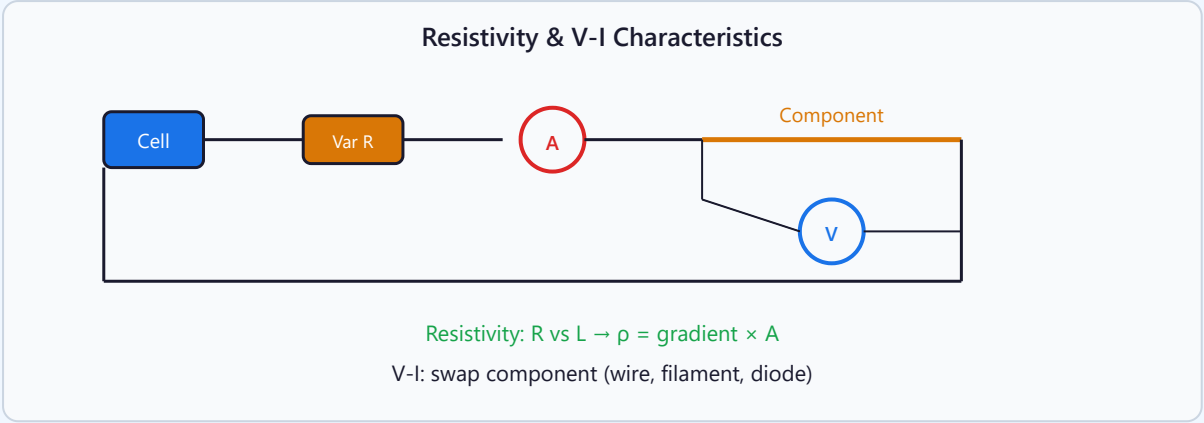
$$f = (1/2L)\sqrt{(T/\mu)}$$

Plot f vs 1/L at constant T: straight line through origin.

Source of Error	Type	Improvement
Judging resonance by eye	Random	Use paper rider on string
L measurement	Random	Measure between nodes, not from vibrator
Mass hanger mass omitted	Systematic	Include hanger in tension calculation
String not uniform	Random	Weigh a measured length to find μ accurately

Exam Tip: At resonance, the amplitude is maximum and the pattern is stable. The number of antinodes equals the harmonic number. Wavelength = 2L/n for the nth harmonic.

⚗ PAG3: Resistivity of a Wire & V-I Characteristics



Apparatus

- **Resistivity:** Constantan wire, ammeter, voltmeter, micrometer, ruler, clips
- **V-I characteristics:** Filament lamp, diode, or resistor; variable resistor or potential divider; ammeter; voltmeter

Method (Resistivity)

1. Measure wire diameter with micrometer (6+ readings). Calculate $A = \pi d^2/4$.
2. Connect circuit with ammeter in series, voltmeter in parallel across the wire.
3. Vary length L using crocodile clips. Record V and I; calculate $R = V/I$.
4. Plot R vs L. Gradient = ρ/A.

Method (V-I Characteristics)

1. Replace wire with the component of interest (filament lamp, diode, or fixed resistor).
2. Use a variable resistor (or potential divider for full range including 0 V).
3. Record V and I for a range of values. Reverse connections for diode.
4. Plot V-I graph.

Safety

- Switch off between readings to prevent overheating
- Filament lamp gets hot — do not touch
- Low voltage (< 6 V)

Analysis

Resistivity: ρ = gradient × A from R vs L graph.

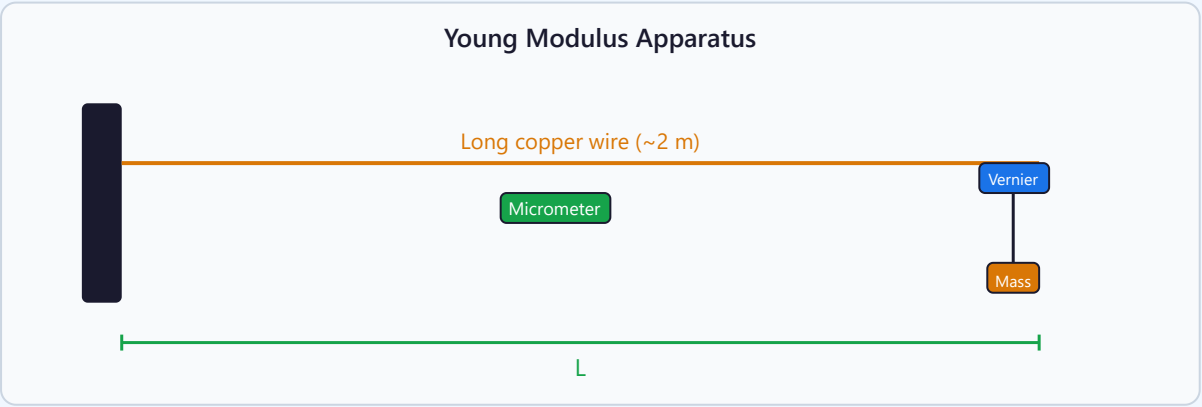
V-I graphs:

- Ohmic conductor (wire): straight line through origin
- Filament lamp: curve showing increasing gradient (resistance increases with temperature)
- Diode: negligible current in reverse; rapid increase above threshold voltage (~0.6 V for silicon)

Source of Error	Type	Improvement
Wire heating	Systematic	Low current; switch off between readings
Contact resistance	Systematic	Clean contacts; firm connections
Parallax on analogue meters	Random	Read at eye level; use digital meters
Diameter variation	Random	Multiple readings at different points

Exam Tip: For V-I characteristics, use a potential divider circuit rather than a simple variable resistor if you need the full voltage range from 0 V upwards. The variable resistor method cannot achieve 0 V across the component.

⚗ PAG4: Young Modulus of a Wire



Apparatus

- Long thin wire (~2 m copper), G-clamp
- Micrometer screw gauge, vernier scale/travelling microscope
- Slotted masses, tape measure, safety goggles

Method

1. Measure diameter at 6+ points with micrometer. Calculate $A = \pi d^2/4$.
2. Measure original length L.
3. Add masses in steps. Record F and extension ΔL from vernier scale.
4. Unload to check elastic behaviour.
5. Plot stress vs strain. Gradient = E (Young modulus).

Safety

- Safety goggles; cushion beneath masses; do not exceed elastic limit

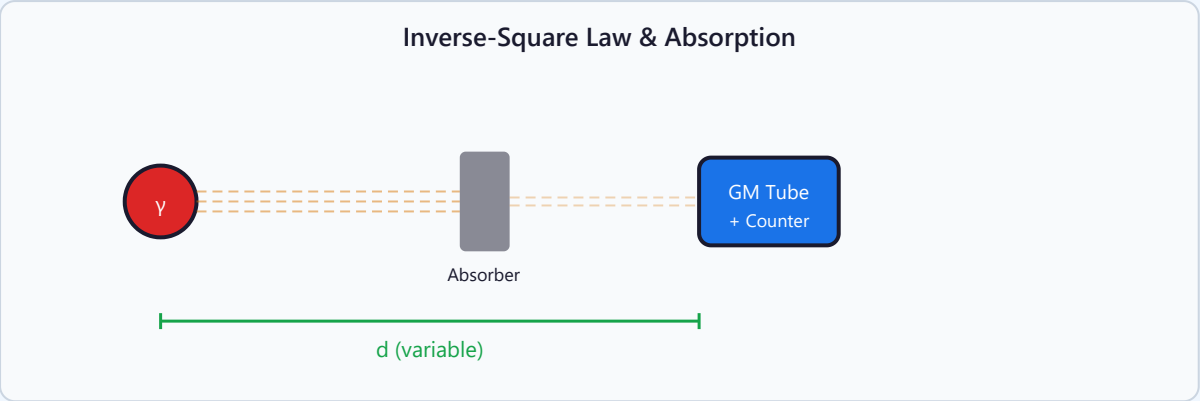
Analysis

$$E = \sigma/\varepsilon = (F/A) \,/\, (\Delta L/L)$$

Source of Error	Type	Improvement
Diameter varies	Random	Multiple readings at different positions
Wire kinks	Systematic	Apply small initial load
Thermal expansion	Systematic	Use reference wire
Small extensions	Random	Use long, thin wire; travelling microscope

Exam Tip: The area under the stress-strain graph up to the elastic limit represents the elastic strain energy per unit volume. The gradient of the linear region gives E. Units: Pa or N m^{-2}.

⚗ PAG5: Inverse-Square Law & Absorption of Radiation



Apparatus

- Gamma source (sealed), GM tube and counter
- Metre ruler, lead absorbers of various thicknesses
- Source holder, clamp

Method (Inverse-Square Law)

1. Measure background count rate (60+ seconds, no source).
2. Place source at distance d from GM tube. Record count rate over 60 s.
3. Repeat for 6+ distances. Take 3 readings at each.
4. Subtract background. Plot corrected count rate vs $1/d^2$.

Method (Absorption)

1. Fix source-detector distance. Measure count rate with no absorber.
2. Insert lead sheets of known thickness. Record count rate for each thickness.
3. Plot ln(corrected count rate) vs thickness. Gradient = $-\mu$.

Safety

- Tongs for source; maximise distance; minimise time
- Point source away from people; lead-lined storage
- ALARA principle

Analysis

Inverse-square: $I \propto 1/d^2$ — C vs $1/d^2$ should be linear through origin.

Absorption: $I = I_0 e^{-\mu x}$ — ln(C) vs x gives gradient = −μ.

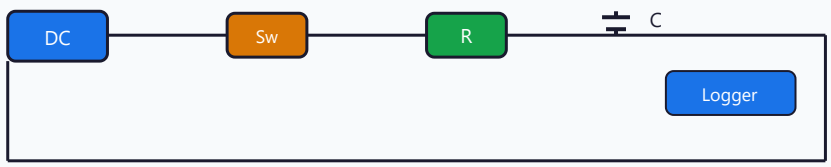
Half-value thickness: $x_{1/2}$ = ln 2 / μ.

Source of Error	Type	Improvement
Random nature of decay	Random	Count for longer; take repeats
Background fluctuations	Random	Long background measurement; multiple readings
Distance to wrong part of tube	Systematic	Measure to GM tube window centre
Absorber thickness not uniform	Systematic	Measure thickness at multiple points

Exam Tip: Always subtract background before any calculations. For the inverse-square law, the source must approximate a point source, which is valid when d is much larger than the source diameter. Plotting ln(C) vs ln(d) should give a gradient of −2.

⚗ PAG6: Capacitor Charge and Discharge

RC Circuit



ln(V) vs t → gradient = −1/RC

Apparatus

- Capacitor (470–2200 μF), resistor, DC supply
- Two-way switch, datalogger or voltmeter + stopwatch

Method

1. Connect RC circuit. Datalogger across C.
2. Charge by connecting supply. Record V vs t.
3. Discharge by disconnecting supply. Record V vs t.
4. Plot ln(V) vs t for discharge. Gradient = −1/RC.

Safety

- Correct polarity; discharge before handling; low voltage

Analysis

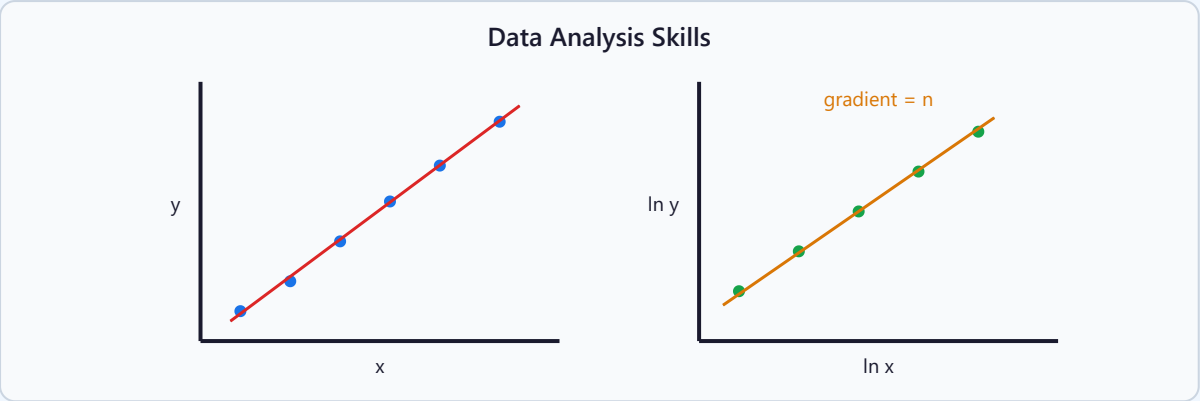
$$V = V_0 e^{-t/RC} \Rightarrow \ln V = \ln V_0 - t/RC$$

τ = RC is the time for V to fall to V_0/e (37%).

Source of Error	Type	Improvement
Voltmeter draws current	Systematic	High-impedance meter or datalogger
Manual timing	Random	Use datalogger
Capacitor leakage	Systematic	High-quality capacitor
Temperature drift	Systematic	Stable environment

Exam Tip: If the ln(V) vs t graph is not straight, this suggests the capacitance or resistance is changing during discharge (e.g. due to temperature effects). Choose R and C for τ = 10–30 s.

⚗ PAG7: Investigation Using Data Analysis



Overview

PAG7 focuses on data analysis techniques rather than a specific physical experiment. It develops skills in processing and analysing experimental data to test relationships and determine unknown quantities.

Key Techniques

- **Linearising relationships:** If $y = kx^n$, plot ln y vs ln x. Gradient = n, intercept = ln k.
- **Log-log graphs:** Used when the power relationship is unknown.
- **Semi-log (ln-linear) graphs:** If $y = Ae^{kx}$, plot ln y vs x. Gradient = k.
- **Calculating uncertainties:** From gradients and intercepts of best-fit and worst-fit lines.
- **Error bars:** Plot on graphs using absolute uncertainties.

Method

1. Collect data from an experiment (often a student-designed investigation).
2. Process the data: calculate derived quantities, convert units.
3. Choose appropriate variables for plotting (to linearise the graph).
4. Plot the graph with error bars.
5. Draw best-fit line and worst-acceptable lines.
6. Determine gradient and intercept with uncertainties.
7. Use the gradient/intercept to determine the physical quantity of interest.

Safety

- Standard laboratory safety for whatever experiment is used

© RAG Learning · Not for redistribution

- Risk assessment specific to the chosen investigation

Analysis

Key formulae for linearisation:

Relationship	Plot	Gradient	Intercept
$y = mx + c$	y vs x	m	c
$y = kx^n$	ln y vs ln x	n	ln k
$y = Ae^{kx}$	ln y vs x	k	ln A
$y = k/x$	y vs 1/x	k	0

Source of Error	Type	Improvement
Anomalous data points	Random	Repeat measurements; identify and exclude outliers
Insufficient data range	Systematic	Collect data across the widest possible range
Poor graph plotting	Systematic	Use appropriate scales; fill at least half the graph paper
Uncertainty in gradient	Random	Use best-fit and worst-fit lines to determine uncertainty

Exam Tip: The uncertainty in the gradient is found from: $\Delta\text{gradient} = (\text{gradient}_{max} - \text{gradient}_{min}) / 2$, where max and min come from the steepest and shallowest lines that pass through all error bars. Always include units in your gradient and intercept.

⚗ PAG8: SHM (Mass-Spring and Pendulum)

SHM Investigations

Spring: T^2 vs m | Pendulum: T^2 vs L

Apparatus

- Spring, slotted masses, clamp stand, stopwatch
- String, bob, protractor, ruler

Method (Mass-Spring)

1. Hang mass m from spring. Pull down slightly, release.
2. Time 10 oscillations. Calculate T. Plot T^2 vs m.
3. Gradient = $4\pi^2/k$.

Method (Pendulum)

1. Measure L from pivot to centre of bob. Displace < 10°, release.
2. Time 10 oscillations. Calculate T. Plot T^2 vs L.
3. Gradient = $4\pi^2/g$, so g = $4\pi^2$/gradient.

Safety

- Stable clamp; clear of swinging masses; do not overstretch spring

Analysis

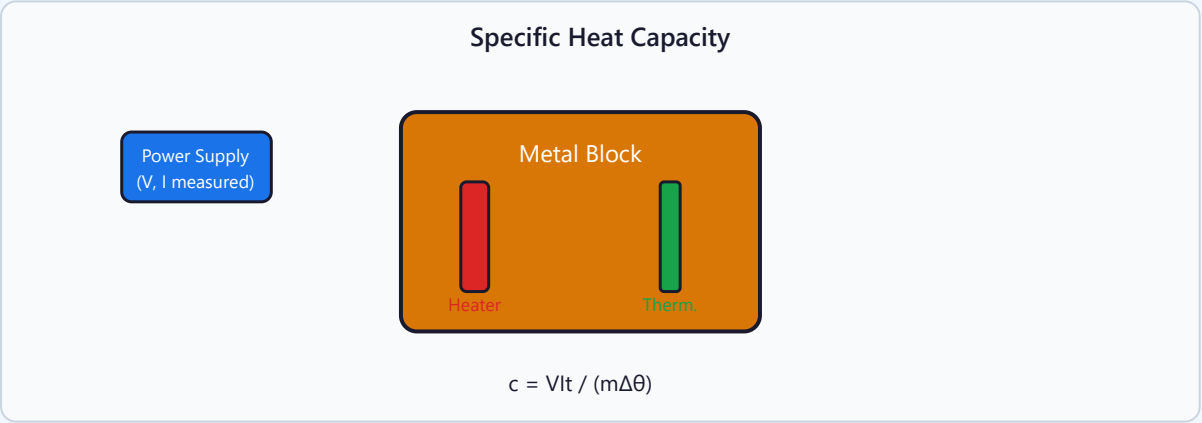
$$\text{Spring: } T^2 = (4\pi^2/k)m \quad \text{Pendulum: } T^2 = (4\pi^2/g)L$$

Source of Error	Type	Improvement

Miscounting oscillations	Random	Use fiducial marker; count from equilibrium
Large amplitude (pendulum)	Systematic	Keep θ < 10°
Non-vertical oscillation (spring)	Random	Release carefully from rest
Reaction time	Random	Time 10+ oscillations

Exam Tip: The pendulum period is independent of mass and (for small angles) of amplitude. The spring period is independent of g, so it would be the same on the Moon. Always plot T^2 (not T) to linearise the graph.

PAG9: Specific Heat Capacity



Apparatus

- Metal block (aluminium/copper) with holes for heater and thermometer
- Immersion heater, thermometer, ammeter, voltmeter
- Power supply, stopwatch, balance, insulation

Method

1. Weigh the block. Insert heater and thermometer (with oil for thermal contact).
2. Record initial temperature. Switch on heater.

3. Record V, I. Heat for a measured time t.
4. Record final temperature. Calculate c = VIt / (mΔθ).

Safety

- Hot block — use tongs; heater fully inserted; heat-resistant mat

Analysis

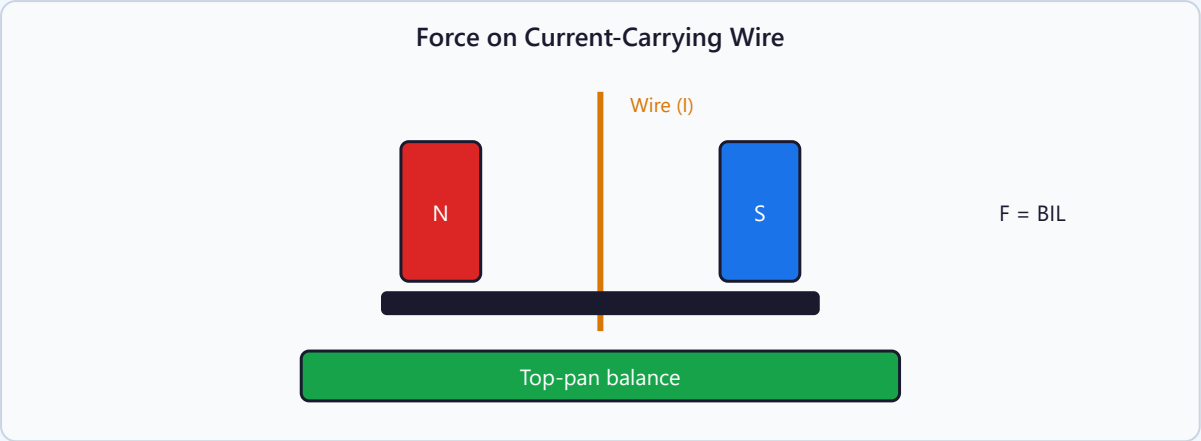
$$c = VIt / (m\Delta\theta)$$

Or plot Δθ vs t: gradient = VI/(mc).

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic (c too high)	Insulate block; cooling correction
Poor thermal contact	Systematic	Use oil in thermometer hole
V/I fluctuations	Random	Monitor throughout; use regulated supply
Thermometer lag	Systematic	Wait for equilibrium

Exam Tip: Measured c is typically higher than the accepted value because of heat losses. A continuous heating method (plotting Δθ vs t) with a cooling correction gives better results than a single measurement.

⚗ PAG10: Force on a Current-Carrying Wire



Apparatus

- Horseshoe/U-shaped magnet on iron yoke
- Copper wire "cradle", top-pan balance (0.01 g)
- DC power supply, ammeter, ruler

Method

1. Place magnet on balance. Zero balance.
2. Position wire in the magnetic field.
3. Vary current. Record I and mass change Δm.
4. F = Δm × g. Measure wire length L in the field.
5. Plot F vs I. Gradient = BL, so B = gradient/L.

Safety

- Low current; short bursts; keep magnets away from electronics

Analysis

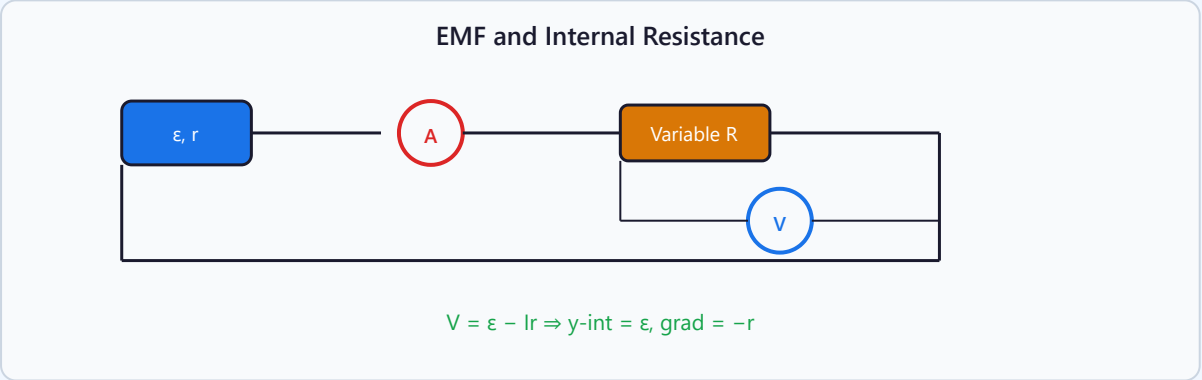
$$F = BIL \Rightarrow \text{plot F vs I, gradient} = BL$$

Source of Error	Type	Improvement
Draughts/vibrations affect balance	Random	Draught shield; stable surface
Non-uniform field	Systematic	Use strong flat-faced magnets

Wire not perpendicular to field	Systematic	Align carefully
Wire heating	Systematic	Short bursts; cooling between readings

Exam Tip: By Newton's third law, the force on the wire equals the force on the magnet. This is why the balance reading changes. F = BIL only applies when wire is perpendicular to B. Use Fleming's left-hand rule for direction.

⚗ PAG11: EMF and Internal Resistance



Apparatus

- Cell, variable resistor, ammeter, voltmeter, switch

Method

1. Connect cell, ammeter, variable resistor in series. Voltmeter across cell.
2. Record V and I for 6+ resistance settings. Switch off between readings.
3. Plot V vs I. y-intercept = ε, gradient = −r.

Safety

- Do not short-circuit cell; switch off between readings

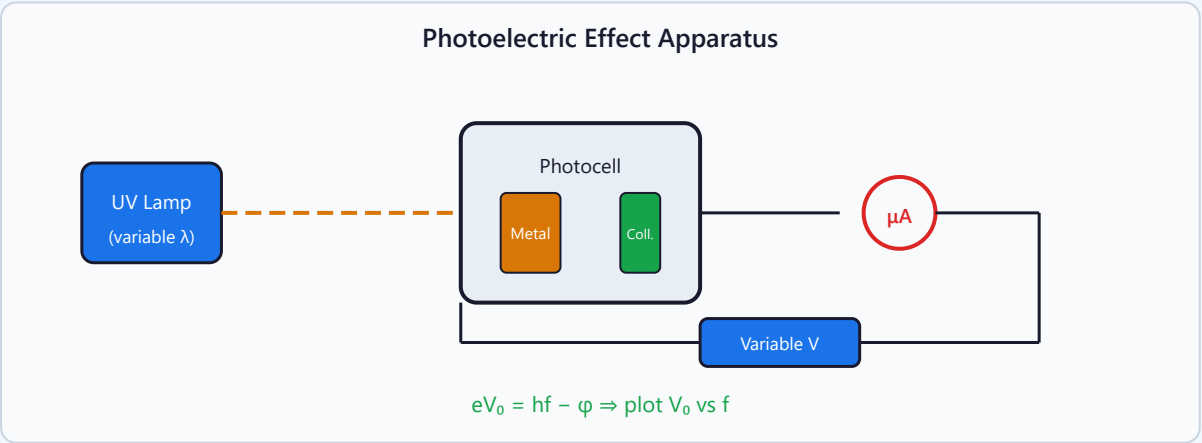
Analysis

$$V = \varepsilon - Ir$$

Source of Error	Type	Improvement
Cell runs down	Systematic	Switch off between readings; work quickly
Meter resistances	Systematic	Digital meters with appropriate impedance
Fluctuating readings	Random	Wait for stability; take repeats
Temperature variation	Systematic	Stable environment

Exam Tip: Never confuse EMF (measured at open circuit, I = 0) with terminal p.d. (measured under load). The "lost volts" = Ir. You can find ε by extrapolating the V vs I line to I = 0.

PAG12: Photoelectric Effect Investigation



Apparatus

- Photocell (vacuum photodiode)
- UV lamp or set of LEDs with known wavelengths / colour filters
- Variable DC power supply (to apply reverse voltage)

- Microammeter (sensitive ammeter)
- Voltmeter

Method

1. Illuminate the metal cathode of the photocell with light of known frequency f.
2. Observe the photocurrent on the microammeter.
3. Gradually increase the reverse (stopping) voltage until the photocurrent drops to zero.
4. Record the stopping voltage V_0 for this frequency.
5. Repeat for at least 5 different frequencies (using filters or different LEDs).
6. Plot V_0 vs f.

Safety

- Do not look directly at UV light — use UV-blocking goggles
- Keep UV exposure to a minimum
- Handle photocell carefully (evacuated glass)

Analysis

From Einstein's photoelectric equation: $eV_0 = hf - \varphi$

$$V_0 = (h/e)f - \varphi/e$$

Plot **V_0 (y-axis) vs f (x-axis)**:

- Gradient = h/e (allows determination of Planck's constant h)
- y-intercept = $-\varphi/e$ (allows determination of work function φ)
- x-intercept = threshold frequency $f_0 = \varphi/h$

Source of Error	Type	Improvement
Stray light reaching photocell	Systematic	Darken the room; use shields around apparatus
Difficult to determine exact V_0 (current drops gradually)	Random	Plot I vs V and extrapolate to I = 0; use sensitive ammeter

Filters not monochromatic	Systematic	Use narrow-band filters or LEDs with known peak wavelength
Thermionic emission from cathode	Systematic	Keep cathode cool; subtract dark current

Exam Tip: The gradient h/e is the same for all metals — only the y-intercept (and hence work function and threshold frequency) changes. Below the threshold frequency, no photoelectrons are emitted regardless of intensity. This is a key piece of evidence for the particle nature of light.

© RAG Learning · Not for redistribution